Work-Based Self-Reflection

Marwan Issa | Computer Science Student | Beirut Arab University

+961 76 488 754 | marwanissa180@gmail.com | Tripoli, Lebanon

Introduction

My work-based learning has developed through the practical projects, technical training, and continuous learning activities I have completed as a Computer Science student. Although my current experience is primarily academic and project-based rather than based on a formal workplace placement, these experiences have allowed me to apply software concepts in realistic problem-solving situations. They have also helped me understand the professional habits required to build reliable, secure, and useful applications.

Applying Knowledge Through Practical Projects

One of the most important parts of my learning has been moving from theoretical concepts to working applications. My SpeedXpo project, a premium car marketplace, gave me experience with full-stack development using PHP, MySQL, Bootstrap 5, JavaScript, AJAX, and PDO. I worked on user authentication, CSRF protection, responsive interfaces, database integration, session management, and CRUD operations. This project helped me see how different technologies must work together rather than being learned in isolation.

My Building Management System and Car Renting System strengthened a different side of my development. Using Java, JOptionPane, Swing, object-oriented programming, event-driven programming, and core data structures, I practiced designing systems around real-world entities and workflows. These projects improved my understanding of how structure, validation, and clear logic affect the reliability of a software solution.

I also completed a Malware Detection project using static analysis and machine learning. Extracting features from Windows PE files and comparing SVM and Decision Tree classifiers required me to think more carefully about data preparation, model evaluation, and measurable results. Working with accuracy, precision, recall, and F1-score showed me the importance of evaluating technical work instead of only making something that runs.

Professional Skills and Workplace Readiness

These experiences have strengthened several professional skills that I will carry into future internships and employment. Problem solving is central to my approach: when an application behaves unexpectedly, I try to isolate the issue, understand the cause, and improve the implementation rather than relying on trial and error alone. I have also developed stronger adaptability because my projects require me to work across different technologies, from Java and PHP to Python and machine-learning tools.

Team communication is another important area of development. My professional goal requires more than technical ability; it requires me to explain ideas clearly, receive feedback, collaborate with others, and remain responsible for my part of a project. I recognize that professional software development is a team activity, so communication and accountability are skills I need to keep strengthening.

> My current learning approach also reflects a professional mindset: after learning a new skill, I try to build a project that applies it. This cycle of learning, building, documenting, reflecting, and improving helps turn theoretical knowledge into practical experience.

What I Have Learned About Myself

My projects have confirmed that I enjoy solving technical problems and learning new technologies quickly. I am especially motivated when a project has a clear practical outcome, such as a web platform, desktop application, or intelligent detection system. I have also learned that I perform better when I break a large task into smaller steps and keep track of progress instead of trying to solve everything at once.

At the same time, I have identified areas that I still need to improve. My Career Development Plan highlights React, Next.js, Node.js, and Express as important next steps, as well as deeper expertise in AI agents. I also need to continue strengthening my GitHub documentation, professional networking, and ability to balance academic responsibilities with independent projects. These are not weaknesses I intend to hide; they are concrete development targets that I can measure over time.

Reflection on Challenges and Growth

A recurring challenge in my learning has been managing time between university coursework, projects, certifications, and personal development. My current plan is to use a weekly schedule, prioritize important tasks, and use techniques such as the Pomodoro method. I also need to control procrastination by setting smaller daily goals and using accountability through friends and online communities. These strategies are important because professional growth depends not only on technical ability but also on consistency.

Another challenge is maintaining progress across many areas of Computer Science. I have explored web development, databases, networking, cybersecurity, data structures, and machine learning. This breadth gives me a strong foundation, but it also means I need to choose priorities carefully. My current direction is becoming clearer: I want to combine web development with AI agents and eventually apply that combination within automotive and racing technology. Having a clearer direction helps me decide which skills deserve the most attention next.

Career Connection

My professional goal is to bridge web development and AI by creating intelligent applications that deliver automated user experiences. In the longer term, I want to become a senior full-stack web developer, gain expertise in AI agents and AI automata, enter the automotive technology industry, and eventually work in racing and Formula 1 technology. This direction connects my technical interests with the kind of fast-paced, engineering-oriented environment I want to experience professionally.

The projects I have completed already provide a starting point for this goal. SpeedXpo connects directly with my interest in automotive technology, while my machine-learning project gives me experience with intelligent systems and evaluation. My next development stage will therefore focus on building stronger full-stack applications, integrating AI capabilities, documenting projects professionally, contributing to open-source work, and growing my professional network.

Key Lessons from My Work-Based Learning

- Practical application matters: understanding a concept becomes stronger when I use it to build something real.
- Security should be considered early: authentication, CSRF protection, API security, and related concepts are part of responsible development.
- Evaluation is essential: technical work should be tested and measured, especially in machine learning.
- Documentation supports professionalism: GitHub repositories, project explanations, and clear communication make technical work easier to understand and share.
- Continuous learning is necessary: my field changes quickly, so professional growth depends on adapting and building new skills.

Professional Development Actions

Action	Why It Matters	Target
Improve React, Next.js, Node.js and Express	Strengthen full-stack capability and modern web development skills.	Next 6 months
Build 2-3 new full-stack projects	Turn new knowledge into measurable practical experience.	6-9 months
Complete current certifications	Formalize learning in AI agents, automata theory and computing fundamentals.	6 months
Improve GitHub documentation	Present projects in a professional and employer-friendly way.	1 month
Secure an internship/job	Apply academic and project learning in a real professional environment.	12 months
Build an AI-integrated web application	Connect my main interests in web development and AI.	Ongoing

Overall, these experiences have moved me from studying individual Computer Science topics to thinking about complete solutions, user needs, security, evaluation, documentation, and continuous improvement. My next step is to turn this foundation into stronger professional experience through internships, advanced projects, certifications, networking, and deeper work at the intersection of web development and AI.